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Possibilities Of Time Travel

How would it be if we could have a cup of coffee with our great grandfather when he was our age? It would be crazy if we got a chance to ride on dinosaurs, how will it be if we can predict natural calamities and save millions. These all activities that seem to be super complicated and full of fiction are not only our imagination, but these all claims also have solid scientific explanation for its possibilities to occur. We can possibly time travel into the future with our existing technology and our understanding of physics till date, the best example is that people living in the International Space Station had already done a time travel of 0.0082 seconds into the future by traveling at such a high speed of approximately 20000 mph compared to earth's rotation.

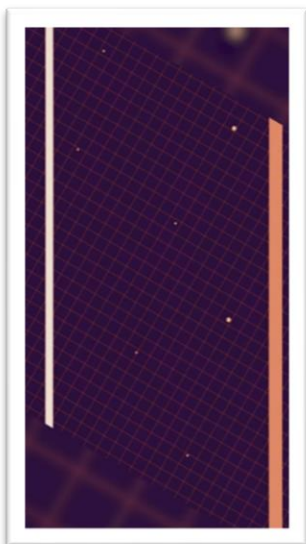
Yes, it is true, time travel is scientifically possible, some methods are travelling through wormholes, by using CTCs (Closed Time like curves), travel at the speed of light or by experiencing strong gravitational force thereby experiencing time dilation, by using cosmic strings. If we discuss about the various possibilities of time travel, we arise with couple of methods through which we can do time travel. So, let us investigate all these methods one by one.

Cosmic String

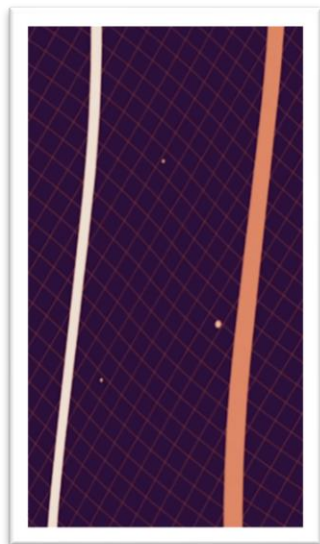
Several physicists believed that our universe is made up of cosmic strings which emerged during Big Bang, these comic strings can be used to facilitate time travel. "We can imagine the

whole universe as made up of cosmic strings, these strings came to existence at the beginning of the universe” (Hawking et al 77). The strings are of very large length extending to the corners of the universe and its width is nearly less than the radius of a nucleus. Imagine we have 2 strings which are parallel to each other like in (fig1.1), these 2 strings travel at very high speed past each other in opposite direction, to visualize it we can consider earth attached to one of the cosmic strings and it could accelerate from 0 mph to about 60 mph in about $1/30^{\text{th}}$ of the second (Hawking et al 77).

Although there is only theoretical existence of cosmic strings no practical evidence is there, “If we encounter with cosmic strings in the universe, they could accelerate us at a very high speed and emerge us at a point where we were in the past” (Hawking et al 77). E.g. If some space traveler was near 2 cosmic strings waiting for its intersection (fig 1.2), as soon as he gets opportunity he kicks on the accelerator and enters into the loop created by distorted space time curvature (fig 1.3), after some time he emerges out into past and he can just say hi to himself in the past (fig 1.4).



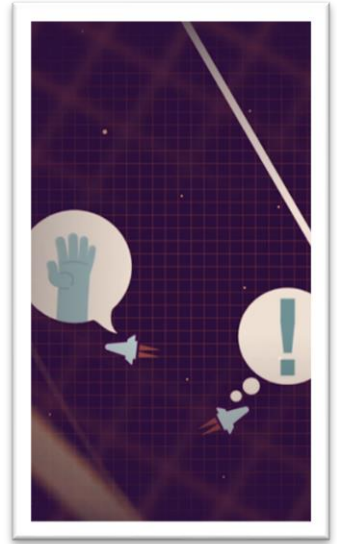
(Fig 1.1)



(Fig 1.2)



(Fig 1.3)



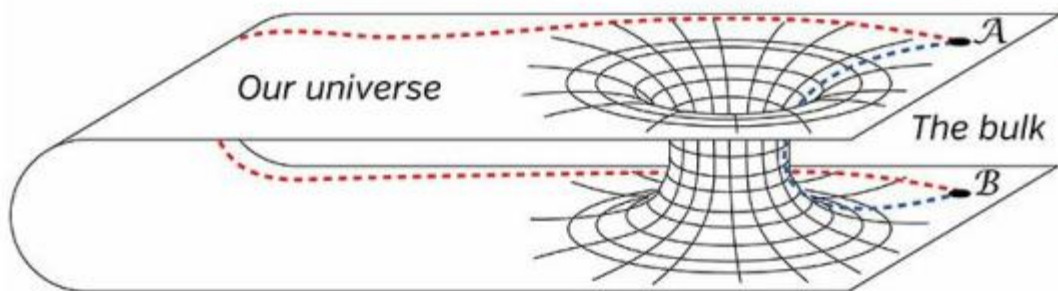
(Fig 1.4)

Wormholes

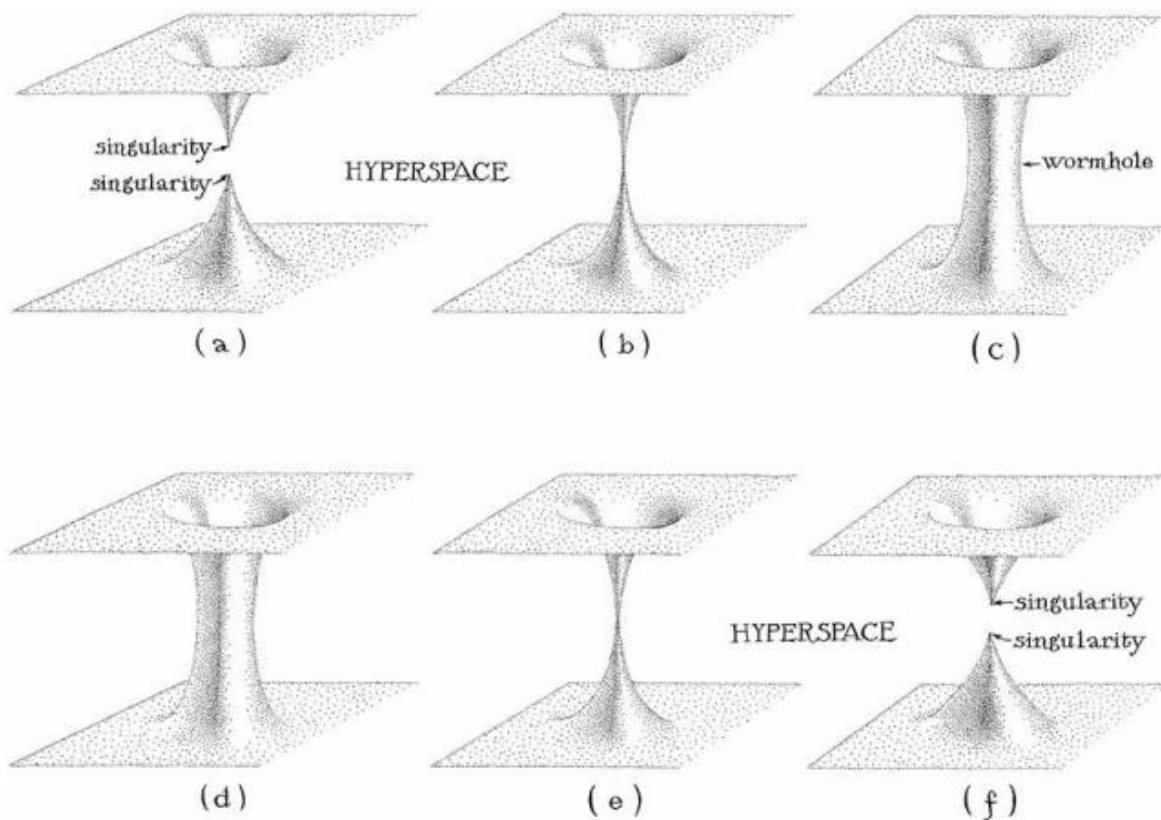
This form of travel has the highest citation and maximum possibility of existence, many scientists agree that this is practically possible. If we forget the scientific part to reduce its complexity, we can understand this method by seeing an example: imagine an ant is moving on an apple, it can get onto other side either by moving whole across the surface of apple or just peeping inside the apple and creating a hole to the other end, this is what wormholes really means in simple sense (Thorne 2014).



We can create a shortcut through space to traverse millions of light years in just fraction of minutes or hours. The below figure is the scientific diagram of wormholes where we can enter from pt A and reach to pt B (Thorne 2014).



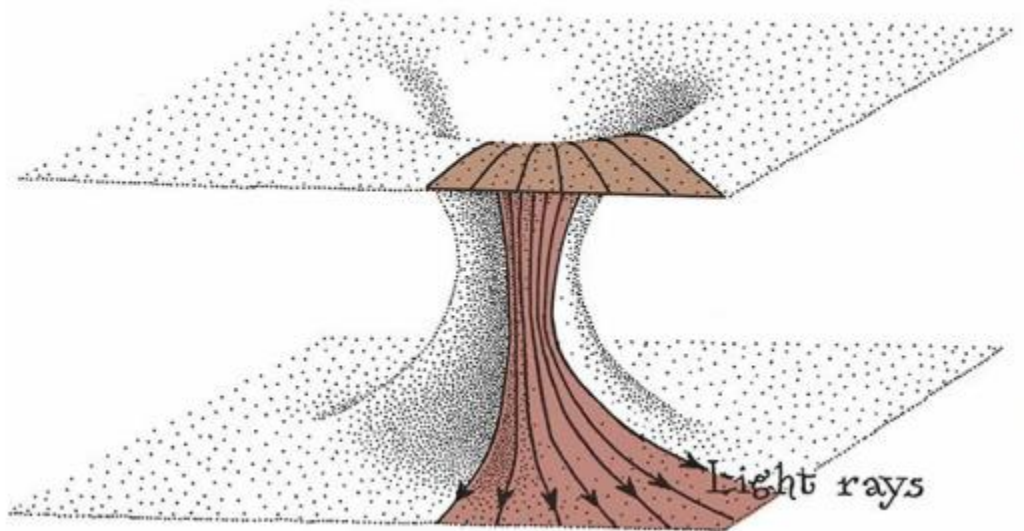
This is called Flamm's Wormhole. Everything looks perfect but it wasn't really, this model of wormhole had one major drawback: "The life cycle of this wormhole is only for fraction of seconds and if someone is struck in between during his transit singularity will just pinch them off and no one can survive" (Thorne 2014) so, we need a traversable wormhole.



After some years of research physicists found solution to this problem keeping the idea of Flamm's wormhole in mind they concluded that we can't directly travel through the center of the wormhole, we have to travel radially inside the wormhole because the tidal gravitational force at the throat is so large if mass of wormhole is greater than 10^4 then the time traveler will be killed who is trying to pass through the throat. (Moris 1987).

Let us take example of light rays to understand how someone can travel through it, rays converge as they move inside and diverge as the come outside. This seems quite simple, but it

isn't, converging is a simple task, but diverging is not possible by normal gravitating bodies like sun or black hole. We need negative mass or relative negative mass for this. "Even if the body don't have this energy, they should look like negative masses relative to the object passing through them" (Hawking et al 1994). To this problem, the idea of negative mass came into rescue, this tedious task of diverging rays or any object is only possible by any celestial body which has negative mass or negative energy. We require a special type of energy called as Exotic matter, there are possibilities that we do not find a negative mass always so any such celestial body should have more negative energy relative to the object passing through it (Thorne 2014).



There is a very positive news that our physicists have already developed fractions of tiny negative mass in laboratories. We can expect that day is very near when we can really do time travel through it. The only existing condition needed to safely pass through traversable wormholes is that it must be threaded with negative energy.

The next possible question will be about methods to find wormholes, these bodies do exist but at very subatomic level in form of Quantum Foam. "There is reason to hope that wormholes do exist naturally on submicroscopic scales in the form of quantum foam" (Thorne

2014). This is basically a hypothesized network of wormholes that is continually fluctuating, and it is of a size of nearly value of planks constant. So, we need to extract this quantum foam and extract it to a size of human then only we can travel through it. Unfortunately, currently we don't have any technology to do this task, but we can expect that humans create some sort of machinery to cater this need (Thorne 2014).

Our doubts regarding the existence of wormholes are definitely genuine, till now there is no practical evidence of its existence, but we have a hope based on our progress in the field of space exploration because even black holes was believed to exist only theoretically for over 100 years until 2017 when its first image was captured by NASA. So, we can believe that it will be discovered someday in the future since it's a theoretically valid concept (NASA 2023).

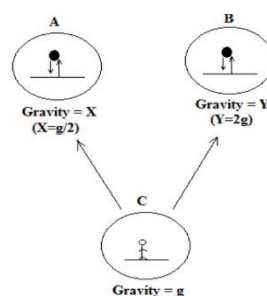
The conclusion to our question regarding the possibilities of time travel through a wormhole is that we can definitely do time travel through it, we have created exotic matter which is needed to keep the mouth of the wormhole open without getting stuck inside but till date we don't have the technology to extract Quantum Foam and enlarge it to a human size to facilitate time travel (Thorne 2014).

Time Dilation

To get a better visualization of this topic we need to first understand what the postulates of Einsteins Theory of Relativity are: first, that the speed of light is constant for all observers; and second, that observers moving at constant speeds should be subject to the same physical laws. "Einstein theorized that time must change according to the speed of a moving object relative to the frame of reference of an observer" (Kanade 2014). This is the fundamental concept of time dilation, time is never considered to be constant, it changes with respect to frame of reference of the observer.

The major question in our mind will be how to believe it, for this scientists conducted one experiment they placed two atomic clocks (most accurate clock): one on an aircraft flying from east to west around the earth and one on earth, when they compared the difference in time among both of these clocks, the clock that was on aircraft at such high speeds was ahead by fraction of seconds from the clock that was on earth (Matson 2024). “It the existent time difference of the elapsed time between two events as checked by the observers” (Einstein et al 2005). This is the technical definition of time dilation as proposed by Einstein through his book *The General Theory of Relativity* which means that he didn’t considered time as a constant entity, but it changes with respect to different frame of refences.

Till this timeline we didn’t consider the factor of gravity, force of gravity have a drastic impact on time dilation, to understand this let us understand by considering an example, in this example we are considering planet A has a gravity of X , planet B has Y gravity and Planet C(observer) has gravity g . Here for this experiment, we are taking the order of gravitational force as $(Y > g > X)$. Now the observer on planet C is observing the ball that is being bounced on both the planets A and B (Kanade 2014).



For Planet A (X):

From the perspective of observer on planet C, the ball will be bouncing at a slower pace in other words since the gravitational force is lesser so the time will pass slowly compared to the planet that observer is living.

For Planet B (Y):

From the perspective of observer C, this is completely contrast for planet B, since gravitational force is much larger than observer's planet, The ball will bounce at a much higher pace. Time will pass fastly on planet B.

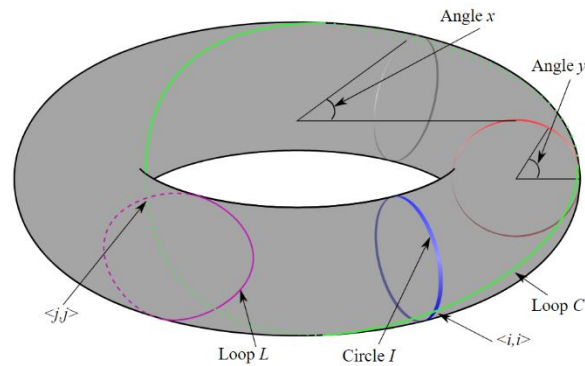
We experience the time difference only while interacting with other planets but at our own place or frame of reference we are completely doing our work effectively. "As per their own perspective, these planets' function or respond to their time standards in an effective manner keeping aside the time dilation factor" (Kanade 2014), in simple words if we are looking at these planets individually everything is going on smoothly there is no time dilation faced by planets individually, we wouldn't feel anything if we are living there it will be a normal life but if you place a video call from someone in planet B to planet A where your father lives, your father on planet A may have reached at your same age since you left planet A.

From the above observation, we conclude that time depends on both perspective and gravity. "Thus, time dilation being the actual difference of elapsed time between two events as measured by observers is the function of 'gravity' along with the 'perspective' factor of the observers" (Einstein et al 2005). From the above quotes it is quite evident that time is heavily influenced by gravity along with frame of reference, time can be modified to any extent depending on the force of gravity. We can experience maximum time dilation if we spend time near a black hole since it has the highest gravitational force.

Closed Time Like Curve

A time-like curve is simply a space-time trajectory where the speed of light is never equaled or exceeded along this trajectory, these curves loop back itself in time making an infinite loop, we can enter through any point and the trajectory is approximately a circular path. If we

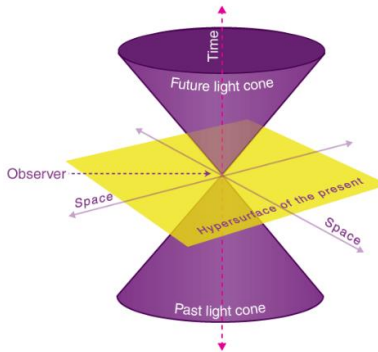
travel through such a curve we would never be required to travel at the speed of light, after certain amount of time one would return to the same point where one visited in the past (Smeenk et al 2023). The below figure represents an appropriate representation,



We can possibly travel through CTC's and emerge at some point in past thereby doing a time travel. There are various space-times which have CTC's well supported by the fundamental equations of general relativity (Smeenk et al 2023). It is proved that there are possibilities of space time fabrics where CTC's can exist which was proved in *Einstein's Theory of Relativity*. Since we have a never-ending universe so at some points, we can find a CTC then we can travel through this loop to emerge out in past.

There is a possibility that CTCs can be a part of wormholes itself with one entry and exit hole .“Space-time, for instance, could have a Minkowski metric everywhere, and yet have CTC's everywhere by having the temporal dimension (topologically) rolled up as a circle or one can have wormhole connections between different parts of space-time which allow one to enter “mouth A” of such a wormhole connection, travel through the wormhole, exit the wormhole at mouth B and re-enter mouth A again (Smeenk et al 2023)” This quote seems super complicated let us understand this from the very fundamental level, first of all we need to understand what

Minkowski Metric is: It is a mathematical model which combines the three dimensions of space and time, it is a way of measuring distance in space-time (Byjus 2023).



The idea here is that even if the spacetime has a Minkowski metric everywhere, closed time like curves (CTCs) could exist. This could happen if the time dimension is topologically rolled up into a circle. In other words, time could loop back on itself in a circular fashion, allowing for closed time like curves (Smeenk et al 2023).

The sentence also introduces the concept of wormholes, which are hypothetical tunnels in spacetime. Wormhole connections could exist between different parts of spacetime. One could enter a wormhole at one point, travel through it, exit at another point, and potentially re-enter the wormhole at the original point (Smeenk et al 2023).

Paradoxes

Over the course of time as we evolved in studying more and more regarding travelling through time many paradoxes emerged to contradict the idea of time travel, we will look into them one by one and find an effective answer to these contradictions. First of all, let us understand is it really possible that someone can change the past? The answer to this question is answered through research papers published by *Stanford Encyclopedia of Philosophy* and *International Journal of Science and Research*. Let us dive into each of them one by one.

The person who is involved in time travel would never be able to change the past some external force will always resist him from doing that. “This means that time travelers can do less than we might have hoped; they cannot right the wrongs of the past; they cannot even stir a speck of dust on a certain day in the past if on the day the speck was unmoved (Smith 2018).” This means that he can’t even touch a speck of dust on that day if that dust remains unmoved the past. It is quite logical if the person remembers that he cleaned the dust, and it remained unmoved then it is not possible that someone changed its position so the time traveler can only be a part of the past and experience the event but can never alter it.

Let us get an answer to grandfather paradox, if someone tries to kill his grandfather then it’s never possible. “Nothing more than such ordinary occurrences is required to stop the time traveler killing grandfather. Hence, backward time travel does not entail the occurrence of impossible events- and so the above objection is defused” (Smith 2018) Any ordinary incident such as the murderer’s gun got jammed, hand got fractured or maybe an apple falls on his head, this is quite evident and proved if that incident didn’t occur in past. Always there will be an external force which will resist you from making any changes to the destiny.

A real life example will make us understand this in a better way, one day you were writing your diary entry and thought why don’t you just take this diary from yourself when you were young so you went to the past to grab that, but you failed because suddenly a dog started barking at you and you were forced to leave that place, you failed and it was destined to be done that way because if you just grabbed that thing in the past then how was it possible that you were writing on that diary before leaving?

What if you just decided to change the past and you are prepared to overcome any failure like wearing a helmet, taking a robot with you to complete that mission, and you even succeeded

in changing the past then the question in our mind will be what next? This controversial question has been answered in *IJSR*'s research paper through the idea of multiverse.

Grandfather Paradox is completely irrelevant based on philosophical grounds. “A multiverse with more than one actual past history does timelessly contain the killing of the grandfather, but only in one of the branches” (Rusho 2022). According to this the author is considering the chance of grandfather getting murdered but only in one universe out of multiple parallel universe, there are many other universes which are running parallel to it and they have covered all the infinite possibilities regarding his arrival following all the physical laws, constants and conditions.

If somebody does time travel, then our universe has explanation to each and every possibility of the events that can occur after the arrival of the person. “Thus, if time travel is physically possible, then there will be an infinitude of branches corresponding to all the possible ways in which time travel could occur. Thus, there will be an infinity of actual events” (Rusho 2022) As soon as the time traveler arrives in the past the universe splits forming many multiverse with infinite possibilities, so maybe at one universe he can kill his grandfather, he can break the leg of his grandfather, he can play with his grandfather, he can have a cup of coffee with his grandfather, or he can take his grandfather along with him for a ride on time machine, there are infinite possibilities, so idea of multiverse is that all these possibilities is addressed in any one out of the many universes.

The only bounding condition is that it will only be true if there is a common root to all the events, it should have a common splitting point then only this event can form multiverse, in this example the splitting point is the arrival of the time traveler (Rusho 2022).

The final answer to all the paradoxes stated above will be that the validity of time travel is not affected through these paradoxes, and time travel is possible if we have a theoretical explanation and technology to do it. Every possible paradox related to this had been explained on philosophical grounds and proved irrelevant both technically and logically.

Conclusion

Time travel is indeed not a topic that should be placed in fiction, this idea is both scientifically and theoretically valid. There exist several diverse and unique methods through which we can do time travel: traversing through a wormhole, experiencing time dilation effect while traveling at speed of light or when near a celestial object which has a very strong gravitational force, entering through closed time-like curves (CTC), by using cosmic strings.

If future generations acquire the ability to do time travel, beyond personal interests, it will create a drastic impact on our societies. Areas like Alaska, Japan, Hawaii are lying on the edges of tectonic plates thereby very much prone to natural calamities, we can do a time travel and predict when will this catastrophe occur and save the lives of maybe millions. This ability will allow historians and scientists to study more deeply regarding our evolution, formation of universe, timeline of catastrophic events, unravel the mysterious facts regarding our planet that got lost with time.

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